THEORY ASSIGNMENT 1

Name: Abhinav Pandey	**Roll No.:** 22515000006
Branch: B.Tech CSE	**Section:** NA(L1)

Q1. HTTP Request–Response Cycle

Ans.

The HTTP Request–Response Cycle is the basic communication process used on the web between a client, usually a web browser, and a web server. HTTP stands for HyperText Transfer Protocol. It allows the browser to request a resource and the server to return the requested information.

Diagram:
Client / Browser → HTTP Request → Web Server → Process Request → HTTP Response → Client / Browser

Major Steps:
1. **Enter URL:** The user enters a website address or clicks a link in the browser.
2. **Send Request:** The browser sends an HTTP request to the appropriate web server. The request contains information about the required resource and the type of operation, such as GET or POST.
3. **Process Request:** The web server receives the request and processes it. If required, it may communicate with an application or database to obtain the requested information.
4. **Prepare Response:** The server prepares an HTTP response. It normally contains a status code, headers and the requested content.
5. **Send Response:** The server sends the response back to the browser through the network.
6. **Display Result:** The browser interprets the received HTML, CSS, JavaScript and other resources and displays the webpage to the user.

Thus, the request–response cycle enables communication between the browser and server and forms the foundation of web communication.

Q2. Introduction to Web Technologies

Ans.

Web technologies are tools, languages, libraries and frameworks used to create, design and operate websites and web applications. The following technologies perform different but related roles:

1. HTML (HyperText Markup Language):
HTML is the standard markup language used to create the structure of a webpage. It defines elements such as headings, paragraphs, links, images, lists, tables and forms. HTML provides the basic content and organization of a webpage.

2. CSS (Cascading Style Sheets):
CSS is used to control the presentation and appearance of HTML elements. It can define colors, fonts, borders, spacing, alignment, layouts and responsive designs. CSS separates the visual design from the webpage structure.

3. JavaScript:
JavaScript is a programming language used to make webpages dynamic and interactive. It can respond to user actions, validate forms, modify webpage content, create animations and communicate with servers without reloading the entire page.

4. Bootstrap:
Bootstrap is a frontend framework that provides ready-made CSS classes, responsive grid layouts and user-interface components. It helps developers create consistent and mobile-friendly webpages with less custom CSS.

5. jQuery:
jQuery is a JavaScript library designed to simplify common JavaScript tasks. It provides easier methods for DOM manipulation, event handling, animations and AJAX requests. It was widely used for making client-side development simpler.

6. React.js:
React.js is a JavaScript library used to develop interactive user interfaces. It encourages developers to divide an interface into reusable components, which makes complex and dynamic frontend applications easier to build and maintain.

Q3. Client-Side vs. Server-Side Technologies

Ans.

Web applications generally use both client-side and server-side technologies. They work together to provide a complete web application.

Client-Side Technologies:
Client-side technologies execute mainly in the user's web browser. Their primary purpose is to create the interface and handle interaction with the user. The browser receives the required files from the server and executes client-side code. Examples include HTML, CSS and JavaScript.

Server-Side Technologies:
Server-side technologies execute on the web server. They are responsible for processing requests, applying business logic, working with databases, authentication and generating responses. Examples include Node.js, PHP, Python/Django and Java/Spring.

Major Differences:
- **Execution:** Client-side runs in the browser; server-side runs on the server.
- **Main purpose:** Client-side handles presentation and user interaction; server-side handles processing and data management.
- **Examples:** HTML/CSS/JavaScript are client-side technologies; Node.js/PHP/Python are common server-side technologies.
- **Database:** Server-side applications commonly communicate with databases, while client-side code normally receives data through server APIs.

Therefore, both sides work together: the client provides the user interface, while the server provides processing and data services.

Q4. Block-Level Elements in HTML

Ans.

Block-level elements are HTML elements that normally start from a new line and take up the available horizontal space of their containing element. They are mainly used to organize content into separate sections or blocks.

Characteristics:
1. They normally begin on a new line.
2. They generally occupy the available width of their parent container.

3. They help divide a webpage into meaningful sections or blocks.
4. They can contain text and, where permitted by HTML rules, other elements.
5. They are useful for creating the overall structure and layout of a webpage.

Examples:
• <div> – general-purpose container.
• <p> – represents a paragraph.
• <h1> – represents a main heading.
• <section> – represents a thematic section of content.
• <header> – represents introductory or navigational content.

Block-level elements are therefore important for organizing webpage content in a clear and structured manner.

Q5. Semantic HTML Elements

Ans.

Semantic HTML elements are elements whose names clearly indicate the meaning and role of the content they contain. Instead of using generic containers for everything, semantic elements give the document a meaningful structure.

Importance in Web Development:
1. **Better Readability:** Semantic tags make HTML code easier for developers to understand.
2. **Accessibility:** Screen readers and other assistive technologies can better understand the structure of the page.
3. **SEO:** Semantic structure helps search engines understand the meaning and organization of webpage content.
4. **Maintainability:** Well-structured HTML is easier to maintain and modify.
5. **Clear Structure:** It clearly separates different parts of a webpage.

Examples:
• <header> – introductory content or header area.
• <nav> – navigation links.
• <main> – main content of the document.
• <section> – a thematic section.
• <article> – independent or self-contained content.
• <footer> – footer information.

Thus, semantic elements improve the structure, accessibility, readability and overall quality of a webpage.

Q6. Creating a Table Using HTML

Ans.

An HTML table is created using the <table> element. The <tr> tag creates a row, <th> creates a heading cell, and <td> creates a data cell. The following code creates the required table:

```
<!DOCTYPE html>
<html>
<head>
  <title>Employee Table</title>
</head>
<body>

  <table border="1">
    <tr>
      <th>Sr. No.</th>
      <th>Name</th>
      <th>Designation</th>
      <th>Department</th>
```

```
    </tr>
    <tr><td>1</td><td>John</td><td>Manager</td><td>Sales</td></tr>
    <tr><td>2</td><td>Smith</td><td>Sr. Manager</td><td>Marketing</td></tr>
    <tr><td>3</td><td>Jane</td><td>Manager</td><td>HR</td></tr>
  </table>

</body>
</html>
```

Q7. Creating a Nested List Using HTML

Ans.

A nested list is a list placed inside another list item. The outer list contains the main item 'River', while the inner unordered list contains the names of rivers:

```
<!DOCTYPE html>
<html>
<body>

  <ul>
    <li>River
      <ul>
        <li>Ganga</li>
        <li>Yamuna</li>
        <li>Brahmaputra</li>
        <li>Narmada</li>
      </ul>
    </li>
  </ul>

</body>
</html>
```

Q8. Types of CSS

Ans.

CSS can be applied to an HTML document in three main ways. Each method has a different use depending on the size and requirements of the webpage.

1. Inline CSS:
In inline CSS, the style is written directly inside the HTML element using the style attribute. It is useful when a particular style needs to be applied to one specific element.
Example: <p style="color: blue;">Hello</p>

2. Internal CSS:
In internal CSS, styles are written inside a <style> tag, normally within the <head> section of the HTML document. It is useful when styles are required for a single webpage.
Example: <style> p { color: blue; } </style>

3. External CSS:
In external CSS, styles are written in a separate .css file and connected to the HTML document using the <link> element. It is the most suitable approach for larger websites because one stylesheet can be reused across multiple pages.
Example: <link rel="stylesheet" href="style.css">

Conclusion: Inline CSS is suitable for individual elements, internal CSS for a single page, and external CSS for reusable styling across multiple webpages.

Q9. Practical Understanding

Ans.

HTML, CSS and JavaScript are three important technologies that work together to develop a modern webpage.

HTML – Structure:
HTML defines what content appears on the webpage and how that content is structured. It creates headings, paragraphs, images, links, tables, forms and other elements.

CSS – Presentation:
CSS defines how the HTML content looks. It controls colors, fonts, sizes, spacing, borders, alignment, page layout and responsive design.

JavaScript – Behavior:
JavaScript adds behavior and functionality to the webpage. It can respond to clicks, validate forms, change content dynamically, perform calculations and communicate with web servers.

Example: In a login page, HTML creates the username and password fields, CSS makes the form attractive and properly arranged, and JavaScript can validate the entered information before submission.

In short: HTML = Structure **|** CSS = Presentation **|** JavaScript = Behavior.

Q10. Short Answer

Ans.

a. What is the role of Bootstrap in web development?
Bootstrap is a frontend framework that provides a collection of ready-to-use CSS classes, responsive grid layouts and user-interface components. It helps developers build websites that work well on different screen sizes, such as desktops, tablets and mobile phones. It also reduces the amount of custom CSS required for common interface elements such as buttons, navigation bars, forms and cards.

b. What is jQuery and why was it widely used in web development?
jQuery is a JavaScript library that provides simpler methods for common client-side tasks. It makes DOM manipulation, event handling, animations and AJAX requests easier to write. It became widely used because developers could perform common JavaScript operations with less code and it helped address differences between browsers. Although modern JavaScript provides many features that reduce the need for jQuery, it remains an important part of web development history.

c. What is React.js and what problem does it solve in frontend development?
React.js is a JavaScript library used to create interactive and dynamic user interfaces. It allows developers to divide a webpage into reusable components, such as buttons, navigation bars, forms and cards. React also provides an efficient approach to updating the user interface when application data changes. This component-based approach makes large frontend applications easier to develop, reuse, test and maintain.